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A parallel finite element scheme for thermo-hydro-mechanical (THM) coupled problems in porous media

Wenqing Wang [a,*], Georg Kosakowski [b], Olaf Kolditz [a,c]

[a] Helmholtz Center for Environmental Research - UFZ, Leipzig, Germany
[b] Paul Scherrer Institut - PSI, Villigen, Switzerland
[c] Technical University of Dresden, Dresden, Germany

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ABSTRACT

Many applied problems in geoscience require knowledge about complex interactions between multiple physical and chemical processes in the sub-surface. As a direct experimental investigation is often not possible, numerical simulation is a common approach. The numerical analysis of coupled thermo-hydro-mechanical (THM) problems is computationally very expensive, and therefore the applicability of existing codes is still limited to simplified problems. In this paper we present a novel implementation of a parallel finite element method (FEM) for the numerical analysis of coupled THM problems in porous media. The computational task of the FEM is partitioned into sub-tasks by a priori domain decomposition. The sub-tasks are assigned to the CPU nodes concurrently. Parallelization is achieved by simultaneously establishing the sub-domain mesh topology, synchronously assembling linear equation systems in sub-domains and obtaining the overall solution with a sub-domain linear solver (parallel BiCGStab method with Jacobi pre-conditioner). The present parallelization method is implemented in an object-oriented way using MPI for inter-processor communication. The parallel code was successfully tested with a 2-D example from the international DECOVALEX benchmarking project. The achieved speed-up for a 3-D extension of the test example on different computers demonstrates the advantage of the present parallel scheme.

1. Introduction

The numerical analysis of multi-field problems in porous media is an important subject for many geo-engineering tasks such as the management of geo-resources (e.g. engineering of geothermal, oil and gas reservoirs) as well as waste management (e.g. chemo-toxic and nuclear waste, CO_2 sequestration) (Doughty and Pruess, 2004; Stephansson et al., 2004; Alonso et al., 2005). For instance, the functional integrity of nuclear waste repositories in the deep geologic underground is influenced by several interfering physical and chemical processes. A typical concept for deep geological disposal of high level radioactive waste is based on a combination of technical and natural barriers. Steel or copper waste containers are placed in emplacement tunnels and the void space is backfilled with bentonite. The emplacement tunnels are located in host rocks like clay or salt that hinder water movement and migration of radionuclides. Due to the depth of the disposal system, considerable overburden loads accompanied by large lithostatic stresses are possible. Moreover, radioactive waste emplacements like spent fuel will generate heat for relatively long times. Therefore, thermo-hydro-mechanical (THM) phenomena are very important to consider in those geological environments (Tsang, 1991; Stephansson et al., 2004). Normally, liquid phase flow is very slow in intact very low-permeable host rocks. However, even in such an environment, vapor flow can develop due to artificially induced thermal gradients caused by heat emitting waste forms (Olivella and Gens, 2000). Gas phases might be produced due to the local heating of the formation water close to hot waste canisters or as a result of chemical reactions during canister corrosion. The mechanical behavior of host rock and buffer materials for waste isolation, such as bentonites, can be considerably altered under variably saturated and non-isothermal conditions. Visco-elasto-plastic material behavior needs to be addressed in the mechanical analysis of engineered barrier/waste systems in salt rock (Wallner et al., 2007). Moisture changes in buffer materials, such as bentonite, results in swelling/shrinking effects which have a large impact to the mechanical stability of the corresponding host-rock/engineered barrier/waste system (Xie et al., 2004; Rutqvist et al., 2005b). In return, thermal and hydraulic processes can cause significant mechanical damage in the near field of the host rock formations. A fully coupled THM analysis is required to gain appropriate understanding of the

* Corresponding author.
E-mail address: wenqing.wang@ufz.de (W. Wang).

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complex process interactions in the near field of chemo-toxic and nuclear waste repositories (Tsang, 1991). Therefore, for the evaluation of possible scenarios, long-term predictions, and safety assessments, numerical modeling tools are increasingly used. In the past decade, many efforts have been done for numerically solving THM problems by different research groups (Rutqvist et al., 2005a; Sanavia et al., 2006; Wang and Kolditz, 2007); especially in the framework of the DECOVALEX (http://www.decovalex.com) and the BAMBUS projects.

From the mathematics point of view, THM processes lead to a coupled initial-boundary-value-problem (IBVP) which needs to be solved numerically. Among the available numerical methods, the finite element method (FEM) is mainly used (Noorishad et al., 1984; Kohl, 1992; Lewis and Schrefler, 1998; Rutqvist et al., 2001a; Ehlers and Bluhm, 2002; de Boer, 2005). Irrespective of the specific numerical method the computation of coupled THM problems is very expensive. This is mainly due to two reasons: degree of freedom (i.e. number of field variables) and strong nonlinearity. There are several ways to improve the computational efficiency, e.g. more efficient numerical algorithms, optimization of memory management in the code, and parallelization techniques. Among them, parallel computing provides the most powerful speed-up. Thanks to the decreasing hardware cost in the past years, the parallel computation is getting very attractive for research in computer science (Salinger et al., 1994; Topping and Khan, 1996; Fujisawa et al., 2003; Shioya and Yagawa, 2005; Tezduyar and Sameh, 2006).

Only little work has been published so far about the parallel finite element analysis of THM coupled problems in porous media. Parallel computations have been conducted previously only on small clusters with up to eight CPU nodes (Wang and Schrefler, 1998; Schrefler et al., 2000; Thomas et al., 2003). Previously developed codes make use of the sub-structuring technique for parallelization. This approach is based on the following steps: (1) divide the system into different sub-structuring levels, which consist of several element cells or super-elements, (2) condense the terms associated with internal nodal points in the sub-structure level assembly, (3) parallel assemble local condensed equation systems at sub-structures, (4) assemble a condensed global equation system with unknowns only associated with sub-structure interface nodes and then solve the problem in a so-called multi-front manner, (5) recover internal unknowns using the condensed equations.

This paper presents a novel parallel scheme which is entirely based on the domain decomposition concept (DDC). To this purpose, the numerical calculation for each individual T/H/M process is partitioned by the a priori decomposition of the finite element mesh into sub-domains. All sub-domain related operations are assigned to the involved CPUs concurrently. The object-oriented parallelization of the finite element scheme for each T/H/M process is realized in three major steps: (1) concurrently establish sub-domain mesh topology for both linear and quadratic interpolations, (2) simultaneously assemble linear equation systems in sub-domains, (3) obtain the overall solution by using sub-domain linear equation solver. For step (3), a Jacobi preconditioned BiCGStab linear solver is implemented. The matrix–vector multiplications, which are the basic computational operations of the solver algorithm, are performed at sub-domain level by all involved CPUs concurrently. Finally, the overall numerical solutions for the entire domain are updated by inter-processor communications. It is important to note that this parallel scheme completely avoids the assembly of a global equation system. All equation assemblies are restricted to sub-domain level.

The paper is structured as follows. The mathematical formulation of the THM problem as well as the corresponding finite element scheme are briefly described in Sections 2 and 3, respectively. Details of the parallelization method are given in Section 4 which also includes some aspects of the object-oriented implementation in the framework of the open-source scientific software GeoSys/RockFlow. The speed-up of the parallelization method is presented together with 2-D and 3-D THM numerical simulations in Section 5.

2. Mechanics of THM coupled processes

The governing equations of coupled THM problems are briefly summarized in this section. The physical processes involved are non-isothermal unsaturated flow, heat transport, and poro-elasto-plastic deformation. The corresponding field variables of the multi-field problem are liquid phase pressure p, temperature T, and displacement vector $\mathbf{u}$. More details on THM mechanics can be found e.g. in Lewis and Schrefler (1998), Rutqvist et al. (2001a), and Wang and Kolditz (2007).

2.1. Non-isothermal unsaturated flow in a deformable porous medium

We apply an extended Richards model for the description of non-isothermal flow in unsaturated deformable porous media. The applicability of the Richards approximation for the selected examples (Section 5) was justified by in previous work (Kolditz and de Jonge, 2004; Wang et al., 2007). According to the extended Darcy law for unsaturated flow (Bear and Bachmat, 1991) the macroscopic liquid phase flow due to pressure gradients, capillarity, and gravity is defined as

$$\mathbf{q}_w = -nS\left(\rho_w \frac{k_{rel}\mathbf{k}}{\mu}(\nabla p - \rho_w \mathbf{g})\right) \quad (1)$$

where n is effective porosity, S is water saturation, ρ_w is water density, k_{rel} is relative permeability function, $\mathbf{k}$ is saturated permeability tensor, μ is water viscosity, p is water pressure (field variable of flow process), $\mathbf{g}$ is gravity acceleration vector.

In addition to classic Darcian processes, vapor flow effects have to be considered in non-isothermal systems. Hereby, we directly adopt the flux and mass balance equations derived by Rutqvist et al. (2001a):

$$\mathbf{q}_v = -D_{pv}\nabla p - f_{Tv}D_{Tv}\nabla T \quad (2)$$

where T is local equilibrium temperature of the porous medium, f_Tv is thermal diffusion enhancement factor. D_{pv} and D_{Tv} are hydraulic and thermal diffusion coefficients,

$$D_{pv} = \frac{D_v \rho_v}{\rho_w RT}$$
$$D_{Tv} = D_v\left(\theta \frac{\partial \rho_{vS}}{\partial T} - \frac{\rho_v p}{\rho_w RT^2}\right) \quad (3)$$

D_v is vapor diffusion coefficient, ρ_v is vapor density, θ is relative humidity defined as

$$\theta = e^{p/\rho_w RT} \quad (4)$$

with the specific gas constant for water vapor R. ρ_{vS} is the saturated vapor density given by an empirical function (Rutqvist et al., 2001a)

$$\rho_{vS} = 10^{-3} e^{19.891 - 4975/T} \quad (5)$$

which is related to vapor density by the expression $\rho_v = \theta\rho_{vS}$.

Darcy flux (1) and, additionally, the vapor flux term (2) are inserted in the fluid mass balance equation, which yields the following field equation for water pressure:

$$
\begin{aligned}
& n\left[\frac{\rho_w-\rho_v}{\rho_w}\frac{\partial S}{\partial p}+S\beta_p+(1-S)\frac{\rho_v}{\rho_w^2 RT}\right]\frac{\partial p}{\partial t} \\
& \quad +\nabla\cdot(\mathbf{q}_w+\mathbf{q}_v)/\rho_w+S\frac{\partial}{\partial t}(\nabla\cdot\mathbf{u}) \\
& n\frac{1-S}{\rho_w}\left(\theta\frac{\partial\rho_{vS}}{\partial T}+\frac{\rho_v P}{RT^2}\right)\frac{\partial T}{\partial t}=0
\end{aligned}
\tag{6}
$$

where β_p is a hydraulic storativity coefficient of the porous medium.

2.2. Poro-thermo-elastic deformation

Deformation processes are described by the momentum balance equation of the porous medium in the terms of stress as

$$\nabla\cdot(\boldsymbol{\sigma}-Sp\mathbf{I}-\alpha E\Delta T\mathbf{I})+\rho\mathbf{g}=0 \tag{7}$$

where $\boldsymbol{\sigma}$ is the effective stress of the porous medium, α is the thermal expansion coefficient, E is the young's modulus, and $\mathbf{I}$ is the identity tensor. The density of the porous medium is composed by three phases, liquid, gas, and solids $\rho=nS\rho_w+n(1-S)\rho_g+(1-n)\rho_s$ with ρ_w, the liquid, ρ_g, the gas and ρ_s, the solid densities, respectively.

2.3. Heat transport in a unsaturated porous medium

The heat flux in an unsaturated porous medium is composed by diffusive and advective parts

$$\mathbf{q}_T=-\kappa\nabla T+nS\rho_w c_w\mathbf{v}T \tag{8}$$

where $\kappa=nS\kappa_w+n(1-S)\kappa_g+(1-n)\kappa_s$ is the heat conductivity of the porous medium. κ_w, κ_g and κ_s are the heat conductivities of water, gas and solid phase, respectively. $\mathbf{v}$ is liquid phase velocity. With the definition of heat flux (8), the governing equation for heat transport in a porous medium is given by

$$c\rho\frac{\partial T}{\partial t}-\nabla\mathbf{q}_T=Q_T \tag{9}$$

where $c\rho=nSc_w\rho_w+n(1-S)c_g\rho_g+(1-n)c_s\rho_s$ is the heat capacity of the porous medium. c_w, c_g, c_s are the specific heat capacities of liquid, gas and solid phases, respectively. Q_T is a heat source term.

3. Finite element formulation

The method of weighted residuals (Lewis and Schrefler, 1998) is applied to derive the weak formulation of the governing equations for water flow (6), deformation (7), and heat transport processes (9) given in Section 2.

Water flow—H process:

$$
\begin{aligned}
& \int_\Omega n\left[(\rho_w-\rho_v)\frac{\partial S}{\partial p}+S\beta_p\rho_w+(1-S)\frac{\rho_v}{\rho_w RT}\right]\frac{\partial p}{\partial t}\omega\,\mathrm{d}\Omega \\
& \quad -\int_\Omega \mathbf{q}_w\cdot\nabla\omega\,\mathrm{d}\Omega+\int_\Gamma \mathbf{q}_w\mathbf{n}\omega\,\mathrm{d}\Gamma \\
& \quad +\int_\Omega \nabla\mathbf{q}_v\omega\,\mathrm{d}\Omega+\int_\Omega S\rho_w\frac{\partial}{\partial t}(\nabla\cdot\mathbf{u})\omega\,\mathrm{d}\Omega \\
& \quad +\int_\Omega n(1-S)\left(\theta\frac{\partial\rho_{vS}}{\partial T}+\frac{\rho_v p}{RT^2}\right)\frac{\partial T}{\partial t}\omega\,\mathrm{d}\Omega=0
\end{aligned}
\tag{10}
$$

Deformation—M process:

$$
\begin{aligned}
& \int_\Omega \frac{1}{2}(\boldsymbol{\sigma}-Sp\mathbf{I}-\alpha E\Delta T I):(\nabla\boldsymbol{\omega}+(\nabla\boldsymbol{\omega})^{\mathrm{T}})\,\mathrm{d}\Omega-\int_\Omega \boldsymbol{\omega}^{\mathrm{T}}\cdot\rho\mathbf{g}\,\mathrm{d}\Omega \\
& \quad -\int_\Gamma \boldsymbol{\omega}^{\mathrm{T}}\cdot\mathbf{t}\,\mathrm{d}\Gamma=0
\end{aligned}
\tag{11}
$$

Heat transport—T process:

$$
\begin{aligned}
& \int_\Omega c\rho\frac{\partial T}{\partial t}\omega\,\mathrm{d}\Omega-\int_\Omega \mathbf{q}_{\mathrm{T}}\cdot\nabla\omega\,\mathrm{d}\Omega \\
& \quad +\int_\Gamma \mathbf{q}_T\cdot\mathbf{n}\omega\,\mathrm{d}\Gamma-\int_\Omega Q_T\omega\,\mathrm{d}\Omega=0
\end{aligned}
\tag{12}
$$

Here $\omega\in\mathscr{V}^1\subset H^1_\Gamma(\Omega)^1$ is the linear test function, and $\boldsymbol{\omega}\in\mathscr{V}^n\subset H^1_\Gamma(\Omega)^n$ is the quadratic test function in n-dimensional space. Ω denotes the model domain with Γ as domain boundary, where boundary conditions have to be specified for all field functions p, T, and $\mathbf{u}$, respectively. A Galerkin method is used to discretize the weak forms (10)–(12) of the THM balance equations. Time discretization is accomplished through implicit finite difference schemes. All variables are approximated by admissible interpolation functions in the Taylor–Hood finite element space. Achieving a reasonable numerical stability and accuracy requires the use of linear interpolation functions for pressure and temperature variables and quadratic interpolation functions for displacement variables (Korsawe et al., 2006).

We consider the staggered method to handle the coupling terms of the balance equations. As a consequence, the coupling terms are treated as right-hand side (RHS) terms. The staggered scheme of the THM coupled problem may be briefly written as

$$
\begin{aligned}
& \mathbf{M}_T\mathbf{T}'+\mathbf{K}_T\mathbf{T}=\mathbf{f}_T(\mathbf{p}) \\
& \mathbf{M}_H\mathbf{p}'+\mathbf{K}_H\mathbf{p}=\mathbf{f}_H(\mathbf{T},\mathbf{u}) \\
& \mathbf{M}_M\mathbf{u}=\int_\Omega \mathbf{B}\,\mathrm{d}\Omega^{\mathrm{T}}\,\mathbb{D}\mathbf{B}\mathbf{u}=\mathbf{f}_M(\mathbf{p},\mathbf{T})
\end{aligned}
\tag{13}
$$

where $\mathbf{T}$, $\mathbf{p}$ and $\mathbf{u}$ are the primary node variables of temperature, water pressure and displacements. $\mathbf{f}$ is the specific RHS term containing the contributions of the coupled processes, $\mathbf{B}$ is the strain displacement matrix, and $\mathbb{D}$ is the elasto-plastic constitutive tensor. $\mathbf{M}$ and $\mathbf{K}$ are process-specific mass and Laplace matrices given by

$$
\begin{aligned}
& \mathbf{M}=\int_\Omega \boldsymbol{N}\mathbb{M}\boldsymbol{N}^{\mathrm{T}}\,\mathrm{d}\Omega \\
& \mathbf{K}=\int_\Omega \nabla\boldsymbol{N}\mathbb{K}(\nabla\boldsymbol{N})^{\mathrm{T}}\,\mathrm{d}\Omega
\end{aligned}
\tag{14}
$$

$\mathbb{M}$ and $\mathbb{K}$ are material tensors. The solution of the THM coupled nonlinear problem (13) is obtained by using a combination of Picard and Newton–Raphson schemes (Wang and Kolditz, 2007). More detailed information on the finite element approach of THM coupled problems can be found e.g. in Lewis and Schrefler (1998).

4. Parallelization method

In this section we describe the parallelization concept for the numerical solution of the THM coupled problem formulated in Eq. (13). Basically, the present parallelization method is based upon the DDC. Domain decomposition is used to split the topological discretization of whole model domain (i.e. finite element mesh) into several sub-domains. Then the finite element contributions are assembled for each of those sub-domains. Finally the sub-domain contributions are reconciled to obtain a solution of the original problem for the whole domain.

In general, the parallelization concept consists of following three basic steps: (1) domain decomposition (Section 4.1), (2) partitioning of global assembly of the algebraic equation systems; i.e. assembly of local sub-domain matrices A_i^k and vectors x_i^k of each process $k=T,H,M$ (Section 4.2), where i is the number of sub-domains, and (3) partitioning of the global linear solver (Section 4.3).

We consider geometric parallelism. This means that all CPU nodes of a parallel machine run the same code for the existing

1: Start
2: Read data including mesh and its sub-domain discretization
3: Construct sub-domain mesh topology for Ω_i
4: Begin THM staggered iteration
    a: Begin nonlinear iteration
        T H M $\Longrightarrow$ Assemble A_i^α and b_i^α, $\alpha = p, T$, or,u
        Compute preconditioning of linear equation
        Solve linear equation
    b: End nonlinear iteration
5: End THM staggered iteration
6: Output results and postprocess

Fig. 1. THM simulation procedure for each processor.

domain decomposition. The major simulation procedure for each CPU node i is summarized into a pseudo-code provided in Fig. 1.

Procedures that were parallelized are the construction of the sub-domain mesh topology, assembly of the sub-domain equation systems, and the linear solver. Actually, the communication among the CPU nodes is only required in the linear solver algorithm.

4.1. Domain decomposition

There are two basic approaches for domain decomposition in use, the Schwarz-method with sub-domains overlapping with each other, and the Schur-complement-method with sub-domains being separated by distinct interfaces (Przemieniecki, 1985; Lions, 1989; Carey, 1989; Farhat and Roux, 1991, 1992). The latter is adopted for the present study in order to assure that the assembly of element matrices and vectors is restricted to sub-domains. The iterative Krylov subspace equation solver only performs local sub-domain matrix–vector multiplications, and, therefore, less inter-processor communication is required. Usually, the domain decomposition is conducted a priori using existing tools such as Jostle (Walshaw et al., 1997; Soper et al., 2004) or METIS (Karypis et al., 1998). These tools minimize the number of border nodes of the sub-domains. This is important for ensuring an optimal load balance and it reduces communication latency during parallel computing (Kemmler et al., 2005).

The domain decomposition procedure is realized as follows:

(1) Partitioning of the global finite element mesh into a number of sub-domains Ω_i. Usually, the number of sub-domains corresponds to the number of available CPU nodes of the parallel machine. Those sub-domain data contain global element indices as well as internal and border nodes of the sub-domain.
(2) Reading of the finite element mesh (MSH file) and of the sub-domain data (DDC file).
(3) Computation of the local finite element topology and local–global mapping relations for each sub-domain Ω_i in a parallel manner. This ensures that the local matrices A_i and vectors x_i can be easily assembled afterwards.

4.2. Partitioning of global assembly

The basic concept of parallelization is to split the global equation system into a number of sub-systems, which are denoted as local equation systems hereafter. This partitioning procedure is conducted according to the domain decomposition (Section 4.1). It is based completely on sub-domain mesh topology. These sub-systems are cast to a number of processors (CPU nodes) concurrently. Applying this technique to the FEM leads to a so-called sub-structuring technique of domain decomposition (Farhat and Roux, 1991, 1992). The most important operation for domain decomposition data is the mapping of global to local data and vice versa. That way, local equation systems are established by assembling matrices and vectors, such as **M**, **K** and **f** for each finite element and each T/H/M process.

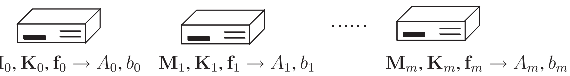

Fig. 2. Parallel assembly scheme of local equation systems.

Consider the element mesh of the entire domain Ω is decomposed into m sub-domains, $\Omega = \Omega_0 \cup \Omega_1 \cup \cdots \cup \Omega_m$ for the THM coupled finite element problem (13). Local equation systems are assembled in these sub-domains with corresponding sub-domain mesh topology and with respect to the initial and boundary conditions. The stiffness matrices A and RHS vectors b are therefore split into blocks related to sub-domain Ω_i

$$
\begin{aligned}
&A_i^p \mathbf{p} = b_i^p \\
&A_i^\mathrm{T} \mathbf{T} = b_i^\mathrm{T} \\
&A_i^u \mathbf{u} = b_i^u, \quad i = 1, \ldots, m
\end{aligned}
\tag{15}
$$

where A_i^p, A_i^T and A_i^u are sub-domain stiffness matrices, b_i^p, b_i^T and b_i^u are sub-domain RHS vectors of fluid flow, heat transport and deformation processes, respectively. The assembly of A_i^p, A_i^T, A_i^u, b_i^p, b_i^T, b_i^u can be easily conducted on different CPU nodes concurrently. The parallel assembly of local equation systems is illustrated in Fig. 2. All required operations, including memory allocations, are managed by each CPU node itself. At this stage inter-processor communication is not yet required.

4.3. Partitioning of global linear solver

The most difficult and challenging part of the parallelization task is the solution of the global equation system. We modified an iterative Krylov solver method for this purpose. Technically, the Krylov method is based on performing a series of products of the system matrix A with several vectors. As a result of the parallelization concept (cf. Section 4.1), these matrix–vector-multiplications can be carried out completely at sub-domain level, i.e. for sub-domain matrix A_i and vectors b_i in the different processors concurrently

$$A_i^k \mathbf{x}_i^k, \quad k = p, T \text{ or } u$$

Important for these local matrix–vector-multiplications is, whether the entries of local matrices and vectors are associated with border nodes or with internal nodes of a sub-domain. If the components of local matrices and vectors are associated with sub-domain border nodes, inter-processor communication is conducted during collection of the norm of $\|A_i^k x_i^k\|$ and during updating the iteration step solutions. Technically, the message passing interface (MPI) function `MPI_Allreduce(...)` is adopted to collect the required data from the different CPU nodes.

In case of sparse linear systems it is possible to accelerate the convergence of the iterative Krylov method by employing a left pre-conditioner M. The modified equation system is given by

$$M^{-1} A x = M^{-1} b \tag{16}$$

We apply a Jacobian pre-conditioner in two steps. At first, the diagonal entries of the local system matrix are computed according to

$$M_i^\alpha = \begin{cases} (a_{kl})_i^\alpha & \text{if } k = l, \\ 0 & \text{if } k \neq l, \end{cases} \quad \alpha = p, T \text{ or } u$$

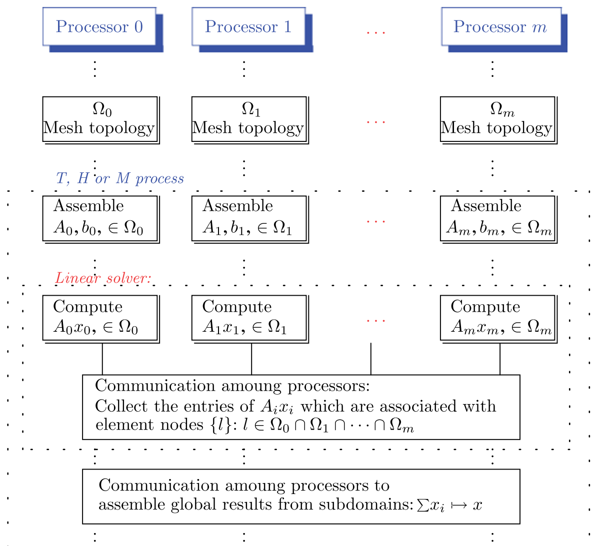

Fig. 3. Schematic of the parallel finite element method.

where $A_i^{\alpha}:=(a_{kl})_i^{\alpha}$. Secondly, the entries which are associated with border nodes of adjacent sub-domains are accumulated. These operations require inter-processor communications.

4.4. Outline of the parallel scheme

The parallelization of the finite element for the solution of THM problems is conducted with MPI. Based on the geometric parallelization concept outlined in Section 4, all CPUs run the same code during program execution. The major simulation steps for each processor (CPU node i) are presented as pseudo-code in Fig. 1. A schematic of the parallel finite element analysis of the THM coupled problems including inter-processor communication is illustrated in Fig. 3. Using the parallelization scheme as described in Sections 4.1–4.3, the assembly of the global system of equations can be completely avoided. The parallelization scheme ensures an efficient load balance between the CPU nodes as long as the domain is well partitioned.

4.5. Accuracy

In this section we prove that the parallel scheme is not loosing accuracy with increasing number of involved CPU nodes. In particular we show that the accuracy of the parallel solution algorithm of the linear equation is not decaying. In the parallelization scheme we have to partition the algebraic equation system (13) and then collect the overall solution in the linear solver scheme. We explain the method and its impact on the accuracy for the case of two partitions. We first assume the algebraic equation can be written in the general form of Eq. (15) $Ax=b$. Our target is to minimize $\|r\| = \|b - Ax\|$ in order to get a solution. When the equation is partitioned into two domains (by local assembly in FEM), we split it into

$$\begin{pmatrix} r_1 \\ r_b^1 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_b^1 \end{pmatrix} - \begin{pmatrix} A_{11} & A_{1b} \\ A_{b1} & A_{bb}^1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_b^1 \end{pmatrix}$$
$$\begin{pmatrix} r_2 \\ r_b^2 \end{pmatrix} = \begin{pmatrix} b_2 \\ b_b^2 \end{pmatrix} - \begin{pmatrix} A_{22} & A_{2b} \\ A_{b2} & A_{bb}^2 \end{pmatrix} \begin{pmatrix} x_2 \\ x_b^2 \end{pmatrix}$$

where $x=(x_1,x_b,x_2)^{\mathrm{T}}$, $b=(b_1,b_b,b_2)^{\mathrm{T}}$ and $r=(r_1,r_b,r_2)^{\mathrm{T}}$ with subscript b denotes border node related entries. The partitioning scheme has to ensure that

$$b_b = b_b^1 + b_1^2$$

Moreover the local domain assembly guarantees the collection of the residual as

$$r_b = b_b^1 + b_b^2 - A_{b1}x_1 - A_{bb}^1 x_b^1 - A_{b2}x_2 - A_{bb}^2 x_b^2$$

Therefore, the accuracy of the solution method does not depend on the number of domain decompositions. However, there may be an accumulation of truncation errors from the iterative solver. This would result in little differences between solutions with different numbers of sub-domains. We show this effect for the 3-D THM example in Section 5.2.3 and demonstrate that the resulting error is very small, normally in the range of the iterative solver tolerance.

4.6. Efficiency

CPU clock time and wall clock time can be used to evaluate the efficiency of a parallel scheme. Since the present parallel scheme is based on geometric parallelism, the overall simulation time is measured when the execution on all computer nodes is finished. The CPU time includes user time for instructions and cache, as well as system time for initiating I/O, paging exceptions and memory allocation. It has a rather low temporal resolution (typically $\frac{1}{100}$ second). CPU time is an appropriate measure to evaluate the performance of the entire program on a shared system. The wall time is the elapsed time, which includes CPU time, communication time and the time needed to release computer or network resources after execution is finished. It should be noted that the wall time is also influenced by specific hardware features, such as speed and bandwidth of the node-to-node connections. The difference between CPU time and wall clock time will vary on different hardware and software platforms. In this paper we use both CPU clock time and wall clock time to evaluate the performance of the present parallel scheme.

4.7. Computer platforms

The computations were performed at two different platforms: CRAY XT3 and SUN X4600 parallel machines. The CRAY XT3 at the Swiss Super Computing Centre (CSCS) is based on AMD x86_64 Opteron CPUs with 2 or 4 GB accessible memory for each CPU node. The entire XT3 compute system consists of 1664 dual core nodes giving 3328 compute CPU nodes, subdivided into 416 compute blades. The whole machine has a peak performance of 17.30 TeraFlops and in total 3.3 TeraBytes of RAM are accessible. Each Opteron processor is directly connected to a dedicated Cray SeaStar chip. Each SeaStar contains a 6-Port router and communications engine and also provides a serial connection to the Cray RAS and management system. The SUN X4600 cluster at UFZ is used for development and testing of parallel codes. It consists of AMD x86_64 based processor nodes, each equipped with 8 x Opteron 885 and 8 GB of RAM per CPU. A total of 80 CPU and 320 GB of RAM give an HPL based peak performance of 238 GFlops. All nodes are connected via low-latency 10 GBit-Infiniband with 1 GBit-Ethernet-Interconnect.

Both platforms are based on x86_64 Opteron nodes, but the node interconnection speed is different. The technical differences of the platforms are important for the achieved speed-up, as we will see in Section 5.2.2.

5. Parallel computation results

As application example for the present parallel FEM scheme we consider the simulation of THM coupled processes in a geotechnical repository system. The present numerical model was developed in the framework of the international code comparison project DECOVALEX-THMC (an acronym for DEvelopment of COupled models and their VALidation against EXperiments) (Barr et al., 2004). This project was dedicated to the analysis of THMC processes (the C stands for geochemical processes) for nuclear waste depository scenarios in different geological environments. The particular "Task-D" benchmark test is based on experimental data obtained in the Grimsel (Switzerland) and Yucca-Mountain (US) test sites. The experiments were aimed at gaining a better understanding of basic processes such as the effect of THM processes on porosity and permeability changes. For this purpose two 2D-test cases have been originally defined (Rutqvist et al., 2008).

The contents of this section is twofold. First, the results of the original 2-D "Task-D" test are given and compared to other codes (Section 5.1). Second, a 3-D extension of the "Task-D" test example demonstrates the efficiency and the limitations of the present parallel scheme for complex THM problems (Section 5.2).

5.1. Verification example (2-D)

5.1.1. Definition

Fig. 4 depicts the definition of the DECOVALEX-THMC "TASK-D" test case as a concept of a buffer system to isolate radioactive waste in crystalline rock. The waste canister is installed into a drift and sealed with bentonite. The drift is embedded in fully saturated rock at a depth of 500 m below ground surface. Since the surrounding crystalline rock is assumed to be fully saturated, water may leak into the initially "dry" bentonite, which leads to swelling of the buffer material. The radioactive waste will heat the bentonite buffer, as well as the surrounding rock for a relatively long time. Therefore, the understanding of the fully coupled THM processes in a long-term period is very important for safety assessment.

The geometry of the 2-D model domain is illustrated in Fig. 4. Material parameters for host rock and bentonite buffer were chosen according to the DECOVALEX TASK D definition (Barr et al., 2004). Particular soil–water-characteristic functions are used for the description of the hydraulic behavior of the partially saturated porous materials. To mimic the heat emission of the radioactive waste, a time-dependent heat power function is applied as thermal source term. Boundary conditions applied for flow, deformation, and heat transport processes are shown in Fig. 4. Further details of the case study definition can be found in Rutqvist et al. (2008).

5.1.2. Results

The 2-D THM simulations have been used for verification and testing of the parallel scheme. We systematically tested the

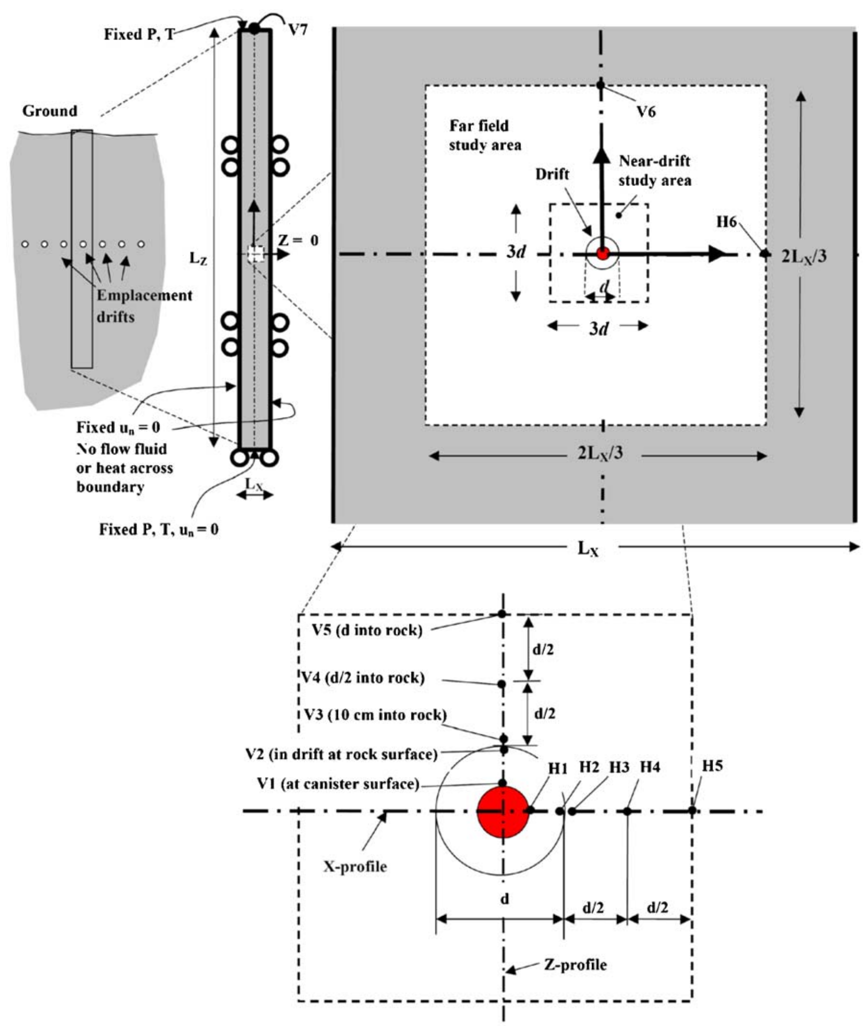

Fig. 4. Definition of the DECOVALEX-THMC "TASK-D" test case (Barr et al., 2004).

individual T/H/M processes in the beginning, and then increased complexity step-by-step by adding the different process couplings. The sequential computation served as a reference solution and was used to test the correct implementation of the parallel scheme (see also Section 5.2.3 for discussion of accuracy). We would like to stress, that we did not find significant differences between the serial and parallel solutions. Within the testing procedure sequential and parallel simulations with 2–20 decomposed domains have been compared.

In addition, we present selected results of the code comparison study that was conducted within the DECOVALEX TASK D. Fig. 5 shows the temperature evolution at selected observation points, a result of the code-to-code comparison between the THM codes of the DECOVALEX project, ROCMAS (Rutqvist et al., 2001a), TOUGH-FLAC (Rutqvist et al., 2001b), THAMES (Chijimatsu et al., 2000), FRT-THM (Liu et al., 2006) and the present parallel code GeoSys/RockFlow.

5.1.3. *Efficiency*

In this section we do a first analysis of the efficiency of the parallel scheme, which in general depends on the following factors: problem size, hardware and software effects.

The speed-up for a 2-D THM simulation based on wall clock time measurements on the SUN X4600 cluster is shown in Fig. 6. We would like to note that the problem size with 1720 nodes and 3152 is rather small. A nearly optimal speed-up can be achieved for up to 16 CPU nodes, then the parallel efficiency goes down. We found that the most important factor is the behavior of the linear solver, which was also reported by Ouarraui and Kaeli (2004). Since there is inter-computer-node communication in the parallel solver algorithm, an increasing number of iterations causes larger communication latency. Therefore, the overall speed-up is affected by the number of solver iterations. The inter-computer-node communication takes place on border mesh nodes during each solver iteration and once on all mesh nodes after solver convergence is achieved. The number of border mesh nodes is monotonically increasing with the number of sub-domains. If the number of border mesh nodes becomes close to the overall number of the mesh nodes, the communication between border nodes will limit the speed-up. In other words, if we decompose a mesh into too many sub-domains, we cannot achieve a further speed-up for the parallel simulation.

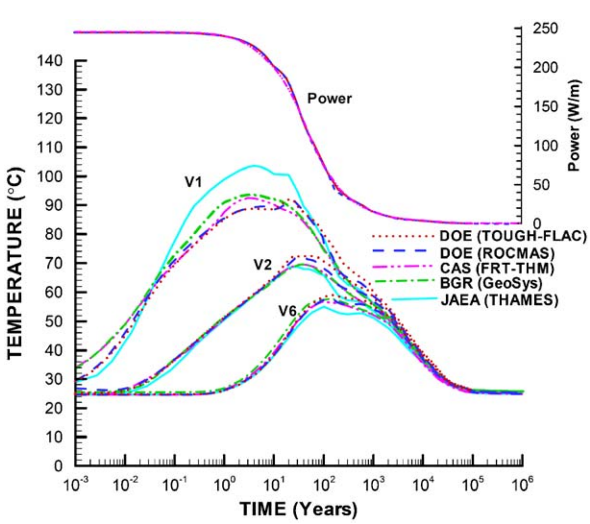

Fig. 5. Temporal evolution of temperature at given observation points (V1: at canister surface, V2: in drift at rock surface, V6: 12.5 m into rock) (Rutqvist et al., 2008).

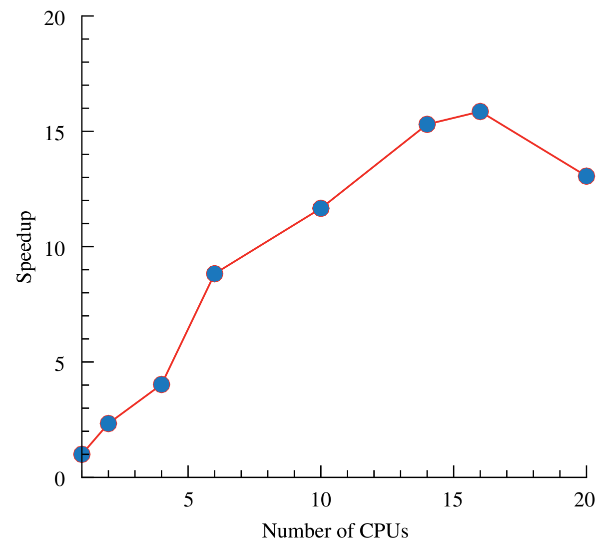

Fig. 6. Speed-up of parallel 2-D THM simulations on a SUN X4600 cluster.

5.2. *Demonstration example (3-D)*

5.2.1. *Definition*

The original DECOVALEX TASK-D problem (Fig. 4) was designed to study long-term effects concerning possible saturation–stress–temperature redistribution resulting from waste deposition. With the help of a 3-D demonstration example we investigate spatial THM processes in the near field of a hypothetical repository. We construct the 3-D model by cutting out a zone of about 100 m from the original 2-D model (Fig. 4). The 2-D vertical domain is then expanded horizontally into the *y*-direction to produce the 3-D model. In order to assign correct boundary conditions in the 3-D model we use results of the 2-D simulation for the corresponding bounding surfaces. The outline and the mesh of the 3-D THM demonstration problem is shown in Fig. 7.

The 3-D domain is discretized into 66 900 tetrahedra with 12 485 element nodes for thermal (T) as well as hydraulic (H) processes (using linear elements) and with 94 215 nodes for the deformation (M) process (using quadratic elements). The mesh is partitioned into 2–100 sub-domains for parallel simulations in order to conduct speed-up measurements. Decomposed domains are depicted in different colors (Fig. 7).

5.2.2. *Results*

Results of the parallel 3-D THM computations are shown in Fig. 8. THM processes are represented by the distributions of the field variables, saturation (non-isothermal unsaturated flow), temperature (heat transport), and stress (thermo-mechanical deformation).

In order to obtain speed-up curves for the parallel scheme, we conducted simulation runs with different numbers of domain decompositions on the CRAY XT3 and SUN X4600 platforms. The technical features are explained in Section 4.7. On the CRAY XT3, the sequential simulation consumed a wall time of 4468 s. The wall time decreased to 132 s for the parallel simulation with 100 CPUs. For all simulations performed on the CRAY XT3, the longest CPU time was identical to the wall clock time. On the newly available SUN X4600, we could use up to 64 CPUs for the parallel simulations. For the sequential reference simulation a wall time of 4598 s was required, which is close to the computation time on the CRAY XT3. Unlike the parallel runs on the CRAY XT3, the wall clock time on the SUN X4600 does not match the longest CPU time.

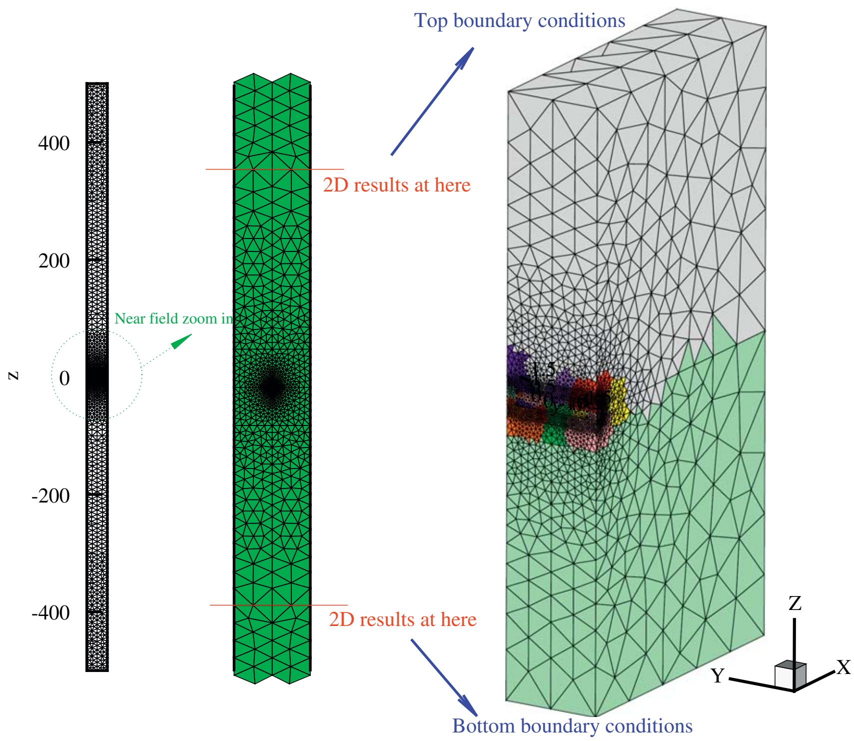

Fig. 7. Mesh of 2-D example (left). Magnified look at 2-D mesh (middle) with positions of trace lines which were used for extraction of boundary conditions for the 3-D THM problem (right).

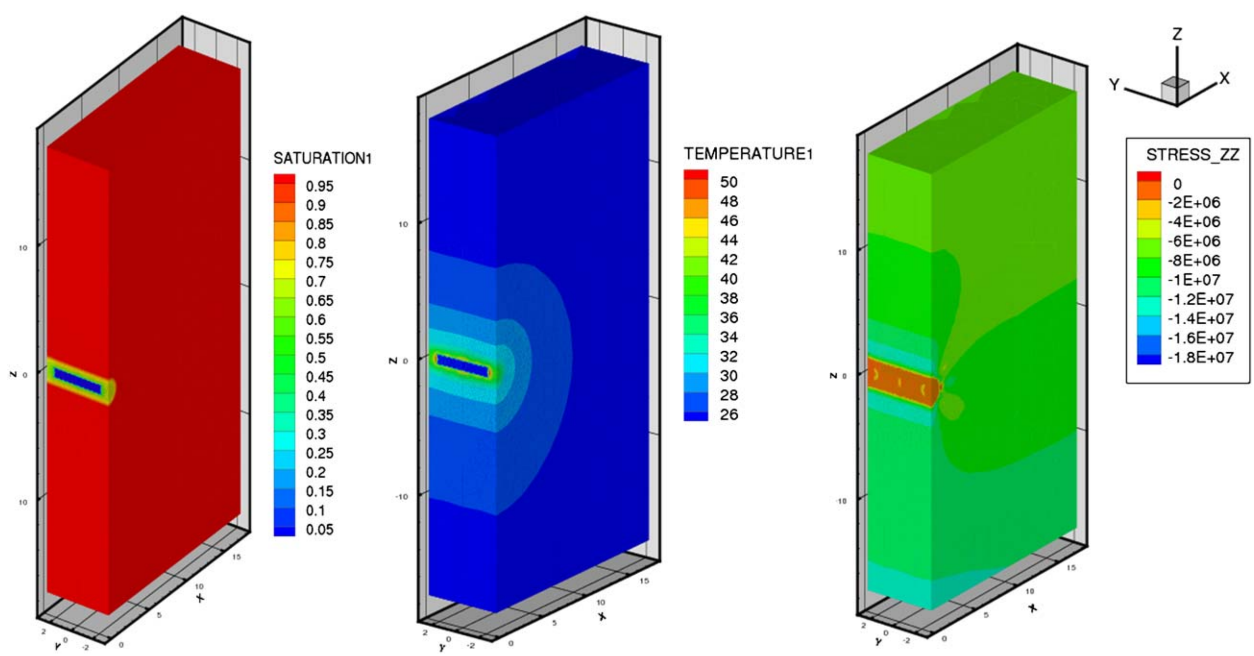

Fig. 8. Computed 3-D distributions of saturation (in m^3/m^3), temperature (in °C) and vertical stress (in Pa) (from left to right).

The speed-up behavior achieved on both platforms is illustrated in Fig. 9. We clearly see that on CRAY XT3 the longest CPU time is identical to the wall clock time. In contrast on SUN X4600, we observe that the deviation between CPU and wall clock times increases monotonically with the number of involved processors. Since we allow I/O operation only on one processor, this time lag is also related to the time duration needed for inter-processor-communication as well as for locating/releasing the communication resources for the parallel computation.

In terms of longest CPU time, the simulations on both platforms give an ideal speed-up of 1:1 for up to 24 CPUs. For more CPUs, the communication latency is gradually increasing which reduces speed-up. Significant deviations between CPU and wall clock time are observed on the X4600 platform beginning from eight CPUs. This "parasitic consumption" is related to the job management system which clearly interferes the processor communication. Along with the increase in the number of involved CPUs, there is a growing delay between wall and CPU

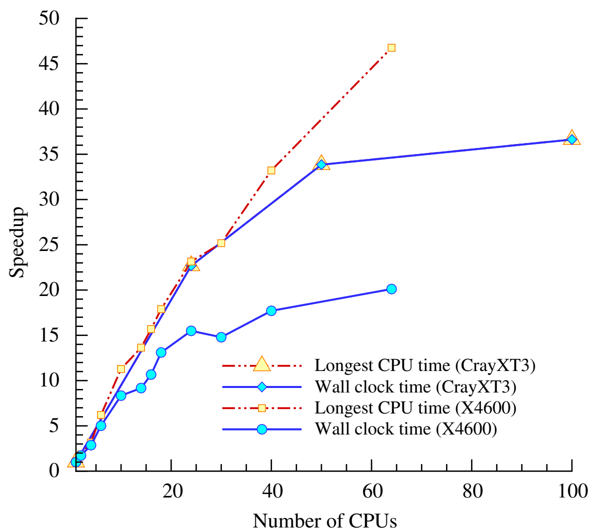

Fig. 9. Speed-up of 3-D THM simulations on CRAY XT3 and SUN X4600 platforms, respectively.

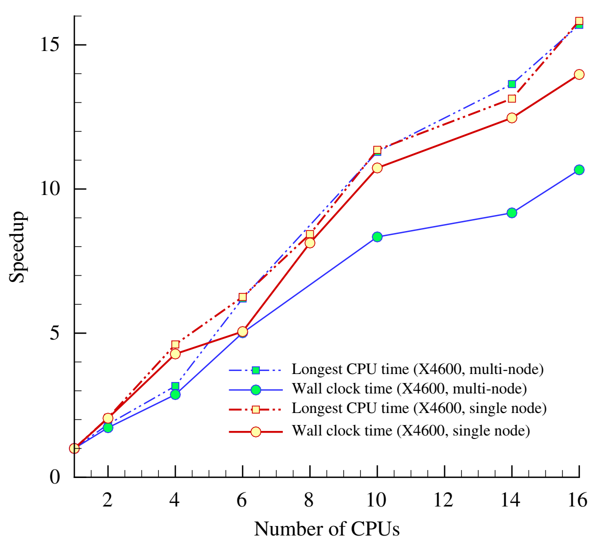

Fig. 10. Speed-up of 3-D THM simulations on SUN X4600 platform with optimized job management.

times. In order to investigate this phenomenon we changed the batch system temporarily in a way to guarantee full access to all eight Dual-Core Opteron CPUs on a processor node of the SUN X4600 cluster. Fig. 10 shows the improvement of the wall clock time if all CPU nodes of a processor node are exclusively used. The discrepancies in wall time behavior shows the importance of comparing speed-up measurements on different platforms for the validation of parallel computations.

5.2.3. Accuracy

Finally, we examine the accuracy of the parallel scheme (according to Section 5.2.3). In all our test cases we did not observe great discrepancies between the serial and parallel solutions. As a typical result we present the accumulated errors for secondary variables. We look at a local grid point where significant changes for the absolute values of the state variables are observed. First, we define the relative difference between solutions x for the serial version (index $=$ 1) and for a parallel version with 40 sub-domains (index $=$ 40) as $e = (x_1 - x_{40})/x_1$. At the end of simulation, the relative difference for the main components of the stress tensor σ_{xx}, σ_{yy}, and σ_{zz} are 8.433×10^{-6}, 4.7×10^{-6} and 5.7×10^{-4}. The difference between serial and parallel versions is in the order of about 10^{-10}, which is in the order of the solver tolerance.

6. Conclusions and outlook

A parallel FEM for the numerical simulation of thermo-hydro-mechanical (THM) coupled processes in porous media has been developed. Although the parallelization technique itself is rather standard, the scheme is (i) new for the parallel computation of this class of coupled multi-field problems (i.e. THM processes in porous media) and (ii) it is analyzed on different medium size platforms (up to 100 CPU nodes) for the first time. For numerical validation, the parallel scheme is applied to the Task-D test example of the DECOVALEX code comparison project.

6.1. Achievements

So far, parallel THM coupled simulations have been conducted only on small clusters with up to eight CPU nodes applying sub-structuring technique (Wang and Schrefler, 1998; Schrefler et al., 2000; Thomas et al., 2003). The present domain decomposition scheme is used for equation assembly and for the linear solver. To our knowledge it is the first time that THM coupled processes in geosciences are solved without any global assembly of the equation system. The iterative solver is preconditioned by a Jacobian method. The sub-domain matrices and vectors are directly used to compute matrix–vector-multiplications and to obtain the overall solution of the linear equation system.

The computational speed-up was analyzed for 2-D and, in particular, 3-D test examples on two different platforms, a CRAY XT3 and a cluster of SUN X4600 machines. The results are summarized in Fig. 9. An examination of the different hardware platforms clearly shows the importance of inter-computer-node communication. There is almost no difference between CPU and wall clock times on CRAY XT3 highlighting the fast connections between all CPU nodes. For the SUN X4600 cluster we see an increasing delay between CPU and wall clock times due to the hardware structure. There is a fast interconnection between the CPUs on each processor node but a slower communication between the processor nodes.

From a practical viewpoint it is now possible to conduct 3-D THM simulations on a 100 CPU-node configuration about 36 times faster than before with the sequential version. In other words, 36 times more numerical simulations are feasible now in the same time for 3-D analysis of practical THM problems. Only a few publications are available so far that present speed-up curves for parallel simulations of similar multi-field problems, therefore a comprehensive classification of the computational efficiency of our parallel code is difficult. The achieved speed-up of the present scheme is at least comparable, or even better than that by Ouarraui and Kaeli (2004) obtained on a cluster with 16 CPUs.

6.2. Deficiencies and future work

The speed-up curves shown in Fig. 9, however, reveal a problem which needs further investigation. Increasing the number of domain decompositions (CPU nodes) will not really improve the computational efficiency of the present parallel scheme even on optimal platforms with fast interconnection between all CPU nodes. The saturation of speed-up for the selected test case starts at about 50 CPU nodes for the 3-D-example. Of course the breakdown of efficiency will occur later for "bigger" problems, but we will not be able to avoid it once we reach a problem-size-dependent threshold. This decay of parallel efficiency is caused by the increase in communication between sub-domain border nodes if the problem is decomposed into too many sub-domains. In order to improve the parallel efficiency further, we will focus on (i) shortening the inter-computer-node communication time by introducing non-blocking communication and on (ii) improving the data exchange scheme on border nodes.

An important lesson learned from this study is, that software and hardware compatibility is very important for the computational efficiency of parallel schemes. In a few years Peta computing (i.e. availability of more than 10^4 CPU nodes) is envisaged. In order to be able to use this supercomputing technology we have to further increase the scalability of the parallel codes for the solution of coupled multi-field problems.

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